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Supplemental Information

Cohort-based long-term ozone exposure-associated mortality risks with adjusted metrics: A systematic review and meta-analysis

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SUPPLEMENTARY MATERIALS

Cohort-based long-term ozone exposure-associated mortality risks with adjusted metrics: A systematic review and meta-analysis

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A total of 27 pages with 5 sections of texts, 8 tables and 7 figures.

Supplementary Texts

S1. Imputation procedures for exposure distribution

To reproduce the distribution, we first extracted the arithmetic means and standard deviations (σ) reported by literatures included for meta-analysis; if unavailable, we estimated the arithmetic means and standard deviations based on the reported descriptive statistics including median, first- and third-quartile, and all the other percentiles; and finally identified the corresponding distribution by Gaussian normal distribution assumption. Reported values were always treated as priority when multiple distribution parameters were given; and we also compared the reported values and our speculated values to check the credibility of our method.

During the process of exposure distribution reproducing, Gaussian normal distribution was presumed prior to log-normal distribution unless violated severely. The centric level, the arithmetic means and medians, were treated as exchangeable, but the arithmetic means were preferred. Theoretically, the minimum and maximum values of the distribution were not deductible, and thus 1st and 99th percentiles were used as proxies. Calculations for σ from key percentiles followed: 75th%ile = mean + 0.6745 σ, 95th%ile = mean + 1.6449 σ, and 99th%ile = mean + 2.3263 σ. If IQRs were stated then IQR = 1.3490 σ; if the 5–95th percentile ranges were reported then range$_{5\text{-}95}$ = 3.2898 σ; if full minimum-maximum ranges were given then range = 4.6527 σ. If more than one distribution features were provided, IQRs were more preferred for σ estimation for higher robustness.

The resampled distributions of O_3 exposure were listed in Table A for original reported metrics and in Table B for harmonised metric into 6mDMA8. By comparing the reported percentiles with distribution-based reproduced estimation values, reasonability of the distribution resampling was verified (Figure A) as the fitting R^2 = 0.84, RMSE = 2.78 ppbV. Theoretically, the minimum and maximum values of the distribution were not deductible, and thus 1st and 99th percentiles were used as surrogates.

Table A. Originally reported and statistically reproduced long-term O_3 exposure distributions statistics for all meta-analysis included studies. The originally reported long-term O_3 exposure levels from literatures were marked in **Bold** font, and the statistically reproduced exposure levels were denoted in brackets. All the exposure concentrations were uniformly scaled into ppbV. Stafoggia et al. 2022[1] did not provide any statistical information of the exposure levels, and thus was not considered.

Study	Metric (ppbV)	Mean	STD	Min	5%	25%	Median	75%	95%	Max	IQR	Range
Abbey et al. 1999[2]	ADA24	**26.1**	**7.7**	(8.3)	(13.5)	(21.0)	(26.1)	(31.3)	(38.7)	(43.9)	**12.0** (10.3)	**43.9** (35.6)
Lipfert et al. 2006[3]	ADMA1	**84.2**	**10.2**	**38.5** (60.5)	(67.4)	(77.3)	(84.2)	(91.1)	(101)	**112** (108)	(13.8)	(73.5)
	ADA24	**54.6**	**6.5**	**24.3** (39.5)	(43.9)	(50.2)	(54.6)	(59.0)	(65.3)	**72.1** (69.7)	(8.8)	(47.8)
Jerrett et al. 2009[4]	6mDMA1											
Krewski et al. 2009[5]	6mDMA1	(60.5)[1]	(15.2)	**33.3**	(35.5)	(50.3)	(60.5)	(70.7)	(85.5)	**104**	(20.5)	**70.7**
Smith et al. 2009[6]	6mDMA1											
Lipsett et al. 2011[7]	ADMA8	**48.1**	**8.7**	**25.4**	(33.8)	(42.2)	(48.1)	(54.0)	(62.5)	**82.6**	**11.2**	**57.2**
Zanobetti et al. 2011[8]	6mDMA8	**46.0**	5.2	**26.6**[2]	(40.2)	**44.2**	**48.6**	**51.2**	(52.6)	**71.4**	**7.0**	**44.8**
Carey et al. 2013[9]	ADA24	**26.4**	**1.2**	**22.7** (23.6)	(24.4)	(25.6)	(26.4)	(27.2)	(28.4)	**32.1** (29.2)	**1.5** (1.6)	**9.4** (5.6)
Jerrett et al. 2013[10]	ADMA8	**50.4**	**14.6**	**17.1** (16.8)	**28.8** (26.8)	**36.8** (41.0)	**50.8**	**61.0** (60.6)	**74.2** (74.8)	**89.3** (84.8)	**24.2** (19.6)	**72.2** (68.0)
Bentayeb et al. 2015[11]	6mDMA8	**49.5**	**4.9**	(20.4)	(25.5)	(45.7)	**49.0**	(52.3)	(57.1)	(60.4)	**6.3** (6.6)	(40.0)
Crouse et al. 2015[12]	6mDMA8	**39.6**	(7.3)	**10.7** (22.0)	(26.9)	**34.3** (34.1)	**39.0**	**44.2** (43.9)	(51.1)	**60.0** (56.0)	**9.9** (9.8)	(49.3)
Tonne et al. 2016[13]	ADA24	**20.6**	**2.0**	(15.9)	(17.3)	**19.3** (19.4)	**20.7**	**22.0** (22.0)	(24.0)	(25.4)	**2.7**	(9.5)
Turner et al. 2016[14]	ADMA8	**38.2**	**4.0**	**26.7** (28.9)	(31.6)	(35.5)	(38.2)	(40.9)	(44.8)	**59.3** (47.5)	(5.4)	**32.6** (18.6)
Di et al. 2017[15]	6mDMA8	**46.3**	(6.0)	(32.4)	**36.3**	(42.3)	**46.3**	(50.3)	**55.9**	(60.2)	(8.0)	(27.7)
Weichenthal et al. 2017[16]	6mDMA8	(38.2)	(6.6)	**1.0** (22.7)	**27.6** (27.2)	**33.7**	**38.1**	**42.6**	**50.5** (49.0)	**60.5** (53.5)	**9.0**	**59.5** (30.8)
Cakmak et al. 2018[17]	6mDMA8	**39.2**	**6.7**	**0.0** (23.6)	(28.2)	(34.7)	(39.2)	(43.7)	(50.2)	**58.8** (54.7)	(9.0)	**58.8** (31.1)
Hvidtfeldt et al. 2019[18]	ADA24	(28.3)	(2.5)	(22.5)	**22.8**	(26.6)	**28.3**	(29.9)	**31.0**	(34.1)	(3.4)	(11.6)
Lim et al. 2019[19]	ADMA8	**39.0**	**4.6**	**26.8** (28.3)	(31.4)	(35.9)	(39.0)	(42.1)	(46.6)	**56.3** (49.7)	(6.2)	**29.5** (21.4)
	6mDMA8	**46.2**	**7.6**	**29.5** (28.5)	(33.7)	(41.1)	(46.2)	(51.3)	(58.7)	**70.4** (63.9)	(10.3)	**40.9** (35.4)
Kazemiparkouhi et al. 2019[20]	6mDMA1	(55.0)	(7.4)	(37.8)	(42.8)	**50.0**	**55.0**	**60.0**	(67.2)	(72.2)	**10.0**	(34.5)
Paul et al. 2020[21]	6mDMA8	**46.9**	(4.7)	(35.9)	(39.1)	(43.7)	(46.9)	(50.1)	(54.7)	(57.9)	**6.4**	(22.0)
Shi et al. 2021[22]	6mDMA8	**40.2**	**4.8**	**17.9**	**30.5**	**37.5**	**40.9**	**43.3**	**47.2**	**50.0**	**5.8**	**32.1**
Strak et al. 2021[23]	6mDMA8	**43.5**	**4.6**	**18.5**	**36.0**	**40.1**	**44.0**	**47.3**	**49.7**	**58.9**	**7.2**	**40.4**
Yazdi et al. 2021[24]	6mDMA8	**41.9**	(3.9)	**31.9**	(35.5)	**39.4**	**42.5**	**44.7**	(48.3)	**50.0**	**5.3**	**18.1**
Bauwelinck et al. 2022[25]	6mDMA8	**39.5**	(1.6)	**19.8**	**34.9**	**38.3**	**39.5**	**40.5**	**42.6**	**46.4**	**2.2**	**26.7**

§ Jerrett et al. 2009[4] did not report the arithmetic mean and standard deviation directly. The values were deducted by weighted averaging the centric concentrations of 4 exposure intervals on the populations given in Table 1 from the original literature.

⊥ Zanobetti et al. 2011[8] did not report the exposure distribution features directly. The quartiles were extracted from the legends provided in Figure 1 of the original literature.

Table B. Statistically resampled distributions of O_3 exposure levels for each study included for meta-analysis. The distribution features included arithmetic mean, standard deviation (STD), minimum, 5th, 25th, 50th (median), 75th, and 95th percentile, maximum, inter-quartile range (IQR), and full range, based on O_3 exposure concentrations scaled by 6mDMA8 metric in ppbV. Values in **Bold** font represented the statistics reported by literature, while the rest indicated resampled values.

Study	Mean	STD	Min	5%	25%	Median	75%	95%	Max	IQR	Range
Abbey et al. 1999[2]	**50.4**	**14.9**	16.1	26.1	40.6	50.4	60.5	74.8	84.9	**23.2**	**84.9**
Lipfert et al. 2006[3]	**80.1**	**9.7**	**36.6**	64.2	73.5	80.1	86.7	96.1	**106.6**	13.2	69.9
Jerrett et al. 2009[4]											
Krewski et al. 2009[5]	50.1	12.6	**27.5**	30.0	41.6	50.1	58.5	70.7	**86.1**	17.0	**58.5**
Smith et al. 2009[6]											
Lipsett et al. 2011[7]	**55.6**	**10.1**	**29.4**	39.1	48.8	55.6	62.4	72.3	**95.5**	**12.9**	**66.1**
Zanobetti et al. 2011[8]	**45.9**	5.2	**26.6**	**40.1**	**44.0**	**48.4**	**51.1**	**52.5**	**71.2**	**6.9**	**44.7**
Carey et al. 2013[9]	**51.0**	**2.3**	**43.8**	47.2	49.5	51.0	52.6	54.9	**62.0**	**2.9**	**18.1**
Jerrett et al. 2013[10]	**58.3**	**16.9**	**19.8**	**33.3**	**42.5**	**58.7**	**70.5**	**85.8**	**103.2**	**28.0**	**83.5**
Bentayeb et al. 2015[11]	**49.4**	**4.9**	20.3	25.4	45.5	**48.9**	52.1	57.0	60.2	**6.2**	39.9
Crouse et al. 2015[12]	**39.5**	7.3	**10.7**	26.8	**34.2**	**39.0**	**44.0**	51.0	**59.9**	**9.8**	49.1
Tonne et al. 2016[13]	**39.8**	**3.8**	30.7	33.4	**37.3**	**40.0**	**42.5**	46.4	49.0	**5.2**	18.4
Turner et al. 2016[14]	**44.2**	**4.6**	**30.1**	36.5	41.0	44.2	47.3	51.8	**68.6**	6.2	**37.7**
Di et al. 2017[15]	**46.3**	6.0	32.4	**36.3**	42.3	**46.3**	50.3	**55.9**	60.2	8.0	27.7
Weichenthal et al. 2017[16]	38.1	6.6	**1.0**	**27.5**	**33.6**	**38.0**	**42.5**	**50.4**	**60.3**	**9.0**	**59.3**
Cakmak et al. 2018[17]	**39.1**	**6.7**	**0.0**	28.1	34.6	39.1	43.6	50.1	**58.6**	9.0	**58.6**
Hvidtfeldt et al. 2019[18]	54.7	4.9	43.5	**44.0**	51.3	**54.7**	57.8	**59.9**	65.9	6.6	22.4
Kazemiparkouhi et al. 2019[20]	**45.1**	**5.3**	**31.0**	36.3	41.5	45.1	48.7	53.9	**65.1**	7.2	**34.1**
Lim et al. 2019[19]	45.5	6.1	31.3	35.4	**41.4**	**45.5**	**49.7**	55.6	59.8	**8.3**	28.6
Paul et al. 2020[21]	**46.8**	4.7	35.8	39.0	43.6	46.8	49.9	54.6	57.8	**6.4**	22.0
Shi et al. 2021[22]	**40.2**	**4.8**	**17.9**	**30.5**	**37.5**	**40.9**	**43.3**	**47.2**	**50.0**	**5.8**	**32.1**
Strak et al. 2021[23]	**43.5**	**4.6**	**18.5**	**36.0**	**40.1**	**44.0**	**47.3**	**49.7**	**58.9**	**7.2**	**40.4**
Yazdi et al. 2021[24]	**41.9**	3.9	**31.9**	35.5	**39.4**	**42.5**	**44.7**	48.3	**50.0**	**5.3**	**18.1**
Bauwelinck et al. 2022[25]	**39.5**	1.6	**19.8**	**34.9**	**38.3**	**39.5**	**40.5**	**42.6**	**46.4**	**2.2**	**26.7**

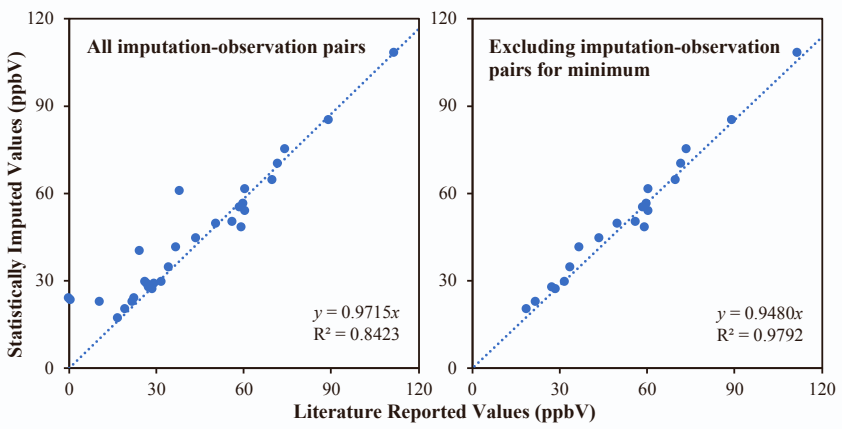


Figure A. Accuracy evaluation of imputed distribution statistics for 25 studies included for systematic review and meta-analysis. The left panel indicated pair-wise accuracies for all distribution statistics including minimum, 5th, 25th, 75th, 95th percentile, and maximum, and the right panel excluded the minimum. Non-intercept linear models were applied to estimate the coefficients of determination (R^2).

S2. Enhanced integrated exposure-response curve-fitting

Exposure-response quantitative relationships were regarded as an additive credit to risk association estimations, as in many cases the adverse health outcomes would respond to exposures in non-linear patterns.[26] We developed the exposure-response non-linear relationships following the integrated exposure-response (IER) curve-fitting approach,[27] based on the mathematical function as:

$$RR(x) = \begin{cases} 1, & x < x_{cf} \\ 1 + \alpha(1 - e^{-\gamma(x - x_{cf})^{\delta}}), & x \geq x_{cf} \end{cases}$$

where the subscript *cf* refers to a counterfactual concentration below which no additional health risks are assumed, exchangeable with the concept of "threshold" or "low concentration cut-off" (LCC).[28]

However, this method required more sufficient sampling of the exposure levels, since small number of studies could only provide limited exposure values to be fitted, which would impair the stability of the curve estimations. In addition, simply drawing an averaged exposure level from a high-quality environmental health study would lose exposure variability information. In this event, we intended to take better advantage of the variabilities in exposure levels by statistically imputing the exposure distributions for each study from the provided statistics mainly including the mean, standard deviation, and percentiles, rather than just using a simple centric averaged exposure concentration, for curve fitting.

To enhance the exposure-response trend construction, we first set the exposure interval nodes by every 5-ppbV ranging between the minimum and maximum exposures reported from all included literatures. In the second step, we selected the studies of which the exposure levels contained each specific node. Thirdly, the counterfactual concentrations were defined as the lowest 5th percentile following a previous research,[28] based on which the cumulative risks at the prescribed nodes were calculated for each study. Fourthly, the individual cumulative risks were pooled into a synthesised risk with 95% UI for each exposure node. As for the fifth step, we generated 10,000 groups of concentration-risk pairs by Monte Carlo random draws according to the pooled risks with 95% UIs, and for each draw the counterfactual concentration was randomly sampled from a uniform distribution between the minimum and 5th lowest percentile concentrations, based on which the draw-specific curve-shape parameters (α, γ, and δ) were estimated. At last, we summarised the distribution patterns of the parameters from all randomised draws, and reproduced the concentration-response curve by another 10,000-time Monte Carlo simulation to characterise the uncertainties. The Monte Carlo simulations were performed in MathWorks® MATLAB 2016b, and the curve fittings were finished in Python 3.8.5 by package Scipy 1.5.2.

S3. Demonstrative procedures of enhanced exposure-response trend curve-fitting

Here is a detailed curve-fitting demonstration for COPD mortality risk by 6mDMA8 metric.

First, identify studies covering each exposure concentration interval level, hereafter called as "*nodes*". According to the resampled exposure distribution, 25.4 ppbV was the lowest 5th%ile concentration reported (Table B), and thence 30 ppbV could be set as the starting node, followed with every 5-ppbV increment as 35, 40, ..., 80 ppbV. For an instance, treating the lowest 5th percentile as the effective lowest exposure level for each study, 5 studies as Abbey et al. (1999)[2], Bentayeb et al. (2015)[11], Crouse et al. (2015)[12], Weichenthal et al. (2017)[16] and Cakmak et al. (2018)[17] covered the minimum node as 30 ppbV by 6mDMA8 metric (Table B), and likewise for the rest nodes. Among the 5 studies identified, only 2 studies as Crouse et al. (2015)[12] and Cakmak et al. (2018)[17] would be used when exploring COPD mortality risks since the rest 3 did not involve COPD studies.

Second, for each exposure node, calculate the study-specific cumulative RRs. Taking Crouse et al. (2015)[12] as an example from the 2 studies covering 30-ppbV exposure, the 5th percentile exposure, 26.8 ppbV by 6mDMA8 metric, was set as the threshold, and hence the cumulative RR for COPD mortality at 30 ppbV exposure should be $e^{(ln0.959\times\frac{30-26.8}{10})}=0.987$, together with its 95% CI: 0.975-0.999. Cumulative RRs for the rest studies and other cause-specific mortalities were calculated in the same way.

Third, pool the study-specific individual risks into overall cumulative RRs at each note by meta-regression within each mortality cause. For COPD mortality risks at 30-ppbV exposure, the cumulative RR by Crouse et al. (2015)[12] was 0.987 (95% CI: 0.975–0.999), and by Cakmak et al. (2018)[17] was 1.000 (95% CI: 0.998–1.002), pooling from which the overall cumulative RR was 0.997 (95% CI:0.990–1.003). The rest intervals followed the same procedure, as summarised in Table C, which could then be applied onto other mortality causes as listed in Table D.

Fourth, for each exposure node, draw a random cumulative risk value based on the normal distribution presumption; and generate a random value between 0 and the lowest 5th percentile across all included studies as the overall threshold level. Then, fit the drawn "node-risk" combinations by the prescribed mathematical formula as presented in Supplementary Text S1, from which three parameters α, δ, and γ could be estimated.

Fifth, duplicate the fourth procedure for 10,000 times with Monte Carlo simulation, and acquire 10,000 groups of the estimated parameter sets, based on which the arithmetic mean, 2.5th and 97.5th percentiles were generated.

Finally, the exposure-response trend curves were reproduced by another Monte Carlo simulation using the estimated 3 parameters together with the uniformly distribution presumed threshold level (between 0 and the cross-study lowest 5th percentile level) to calculate the cumulative RRs through the full range of exposures. The integrated exposure-response curves were plotted in Figure B.

Mathematically speaking, the cumulative risks will flatten at 1 + α, which indicates the anticipated highest risks. For all-cause O_3-associated mortality, the cumulative risk would saturate at around RR = 1.12 by 6mDMA8 metric of O_3 exposure; respiratory mortality at RR = 1.11; cardiovascular mortality at RR = 1.30; and congestive heart failure mortality at RR = 1.68; while COPD mortality risks did not manifest level-off tendency in observed exposure range, as the cumulative risks kept speedily climbing till 80 ppbV.

Large uncertainties were observed in the Monte-Carlo fitted exposure-response curves, which should mainly be ascribed to the limited number of large-scale cohort-based epidemiological studies. We therefore suggest researchers critically scrutinise this methodology and cautiously use our results.

Table C. Cumulative relative risks of chronic obstructive pulmonary disease (COPD) mortality by reproduced O_3 exposure distribution.

Study	30 ppbV	35 ppbV	40 ppbV	45 ppbV	50 ppbV	55 ppbV	60 ppbV
Zanobetti et al. 2011[8]		1.23 (1.13, 1.30)	1.33 (1.18, 1.43)	1.43 (1.23, 1.58)	1.55 (1.29, 1.75)	1.68 (1.35, 1.93)	1.81 (1.41, 2.14)
Crouse et al. 2015[12]	0.99 (0.98, 1.00)	0.96 (0.83, 0.99)	0.93 (0.88, 0.99)	0.90 (0.83, 0.99)	0.87 (0.78, 0.99)		
Turner et al. 2016[14]		1.19 (1.11, 1.28)	1.27 (1.15, 1.41)	1.35 (1.19, 1.55)	1.44 (1.24, 1.71)	1.54 (1.29, 1.88)	
Cakmak et al. 2018[17]	1.00 (0.99, 1.00)	1.00 (0.96, 1.05)	1.00 (0.94, 1.06)	1.00 (0.92, 1.08)	1.00 (0.91, 1.10)		
Kazemiparkouhi et al. 2019[20]		1.16 (1.15, 1.17)	1.22 (1.21, 1.24)	1.30 (1.28, 1.32)	1.37 (1.35, 1.40)		
Lim et al. 2019[19]		1.09 (1.02, 1.19)	1.13 (1.02, 1.27)	1.17 (1.03, 1.35)	1.21 (1.03, 1.44)	1.25 (1.04, 1.54)	
Heterogeneity I^2	65.3%	94.7%	94.7%	94.7%	94.7%	58.7%	-
Pooled RRs	**1.00 (0.99, 1.00)**	**1.09 (1.01, 1.19)**	**1.13 (1.01, 1.27)**	**1.17 (1.02, 1.35)**	**1.21 (1.02, 1.44)**	**1.48 (1.25, 1.76)**	**1.81 (1.41, 2.14)**

Table D. Relative risks of all-cause, respiratory, cardiovascular and congestive heart failure mortality by distribution-reproduced O_3 exposures.

Concentrations	RR	LB	UB	I^2
All-cause Mortality				
30 ppbV	1.003	1.002	1.003	93.2%
35 ppbV	1.008	1.007	1.009	93.8%
40 ppbV	1.006	1.004	1.008	93.2%
45 ppbV	1.008	1.005	1.010	93.7%
50 ppbV	1.009	1.006	1.012	93.5%
55 ppbV	1.021	1.019	1.023	85.6%
60 ppbV	1.025	1.022	1.027	84.1%
65 ppbV	1.028	1.025	1.030	84.1%
70 ppbV	1.039	1.035	1.043	84.1%
75 ppbV	1.040	1.036	1.043	84.7%
80 ppbV	1.040	1.036	1.043	84.7%
Respiratory Mortality				
30 ppbV	1.088	1.079	1.096	94.5%
35 ppbV	1.090	1.082	1.098	97.0%
40 ppbV	1.094	1.089	1.100	94.3%
45 ppbV	1.094	1.089	1.100	96.0%
50 ppbV	1.096	1.090	1.102	96.1%
55 ppbV	1.099	1.093	1.106	93.6%
60 ppbV	1.098	1.094	1.101	51.7%
65 ppbV	1.098	1.094	1.101	48.7%
70 ppbV	1.098	1.094	1.101	46.4%
75 ppbV	1.093	1.067	1.118	5.5%
Cardiovascular Mortality				
30 ppbV	1.028	0.984	1.071	98.5%
35 ppbV	1.042	1.007	1.079	98.1%
40 ppbV	1.048	1.003	1.096	97.9%
45 ppbV	1.062	1.003	1.125	97.9%
50 ppbV	1.077	1.003	1.154	97.9%
55 ppbV	1.010	0.965	1.057	83.4%
60 ppbV	1.008	0.955	1.063	79.6%
65 ppbV	1.009	0.949	1.072	79.6%
70 ppbV	1.010	0.944	1.081	79.6%
Congestive Heart Failure Mortality				
35 ppbV	1.095	1.051	1.142	84.9%
40 ppbV	1.134	1.071	1.201	84.9%
45 ppbV	1.174	1.091	1.264	84.9%
50 ppbV	1.216	1.112	1.330	84.9%
55 ppbV	1.234	1.030	1.478	87.5%
60 ppbV	1.254	1.122	1.349	-

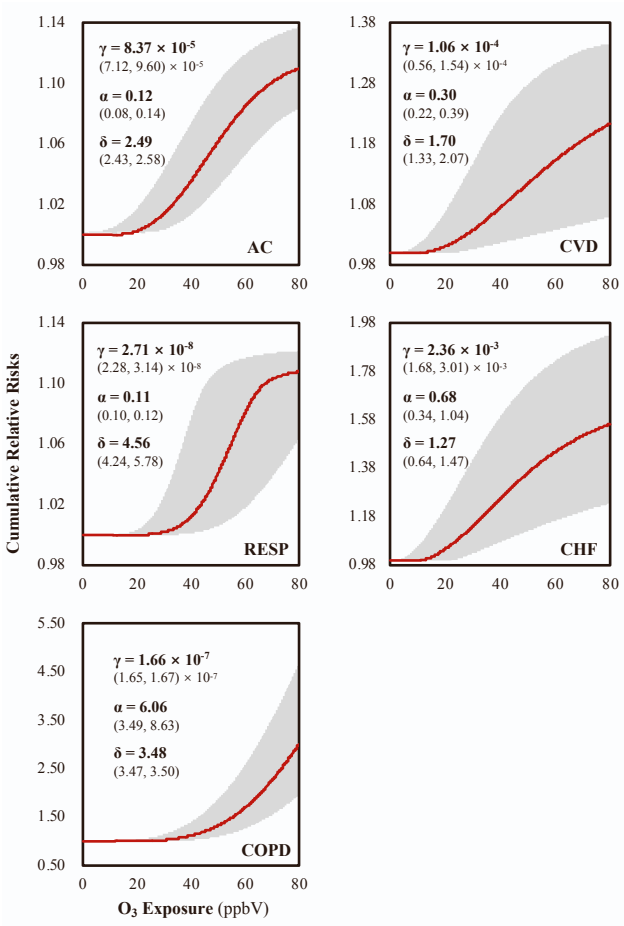


Figure B. Estimated exposure-response curves for mortality risks from all-cause (AC), all respiratory diseases (RESP), chronic obstructive pulmonary disease (COPD), all cardiovascular diseases (CVD), and congestive heart failure (CHF). The trend curves were plotted by centric cumulative relative risk values with 95% uncertainty intervals (95% UI). Estimates of 3 parameters were shown with 95% confidence intervals (95% CI).

S4. Undefined metric imputation

For Carey et al. (2013)[9], the O_3 exposure metric was not stated clearly, but exposure concentrations were reported as 26.4 ± 1.2 ranging between 22.7–32.1 ppbV. Since the cohort study was conducted in UK, we checked the corresponding concentrations spatiotemporal closely from TOAR observations, and we found by ADA24 metric the average O_3 level was 25.7 ± 8.3 ppbV, with IQR 19.8–31.3 ppbV. We thus deduced the metric should be ADA24.

For Jerrett et al. (2009)[4], not any O_3 exposure distributions were reported, but only the minimum-maximum range was given as 33.3–104.0 ppbV. We calculated the average level by our own, as weighting the population with interval centric point: 43.2 ppbV by 126 206, 55.3 ppbV by 95 740, 59.95 ppbV by 106 545, and 83.25 ppbV by 120 359, to obtain the estimated average as 60.5 ppbV. According to the US observations by TOAR, the average levels were around 62.8 ppbV by warm-season DMA1 and 50.9 by full-year DMA1, the former of which was closer to the literature-generated value, 60.5 ppbV, and we thus inferred the exposure metric should be 6mDMA1 more reasonably.

Jerrett et al. (2013)[10] did not report which metric they used, but detailed distribution patterns were provided. Based on TOAR observations by ADMA8 in US, O_3 concentrations were 54.1 ± 16.1 (IQR: 42.4–65.5, range: 17.6–102.5) ppbV, closer to the reported values as 50.4 ± 14.6 (IQR: 36.8–61.0, range: 17.1–89.3) ppbV than 6mDMA8 which was of average levels as 63.1 ± 18.2 ppbV.

S5. Interpretation and procedure of cross-metric linear conversion

For direct cross-metric linear conversion, taking conversion from ADA24 to ADMA8 as an interpretation example, the prediction could be approximated by ADMA8 = 1.67 × ADA24 (R^2 = 0.97, **Figure 2**), which means each 10-ppbV change by ADA24 metric should correspond to 16.7-ppbV change by ADMA8 metric, and likewise for the rest conversions. The conversion factors deviant from 1 revealed that unit changes in O_3 concentrations by different metrics could be of discrepant realistic implications, and hence unifications of O_3 exposure metrics should be necessary prior to meta-analysis.

For cross-metric relative risk conversion, taking Smith et al. (2009)[6] as an example, the originally reported all respiratory mortality HR = 1.0057 (95% CI: 1.0020–1.0094) by every 1-μg/m^3 incremental O_3 exposure by 6mMDA1, equally HR = 1.0585 (95% CI: 1.0202–1.0981) by every 10-μg/m^3 (equal to 10/1.96 = 5.10-ppbV) incremental O_3 exposure. Converting μg/m^3 into ppbV, every 10-ppbV increasing O_3 exposure by 6mMDA1 should correspond to HR = 1.1179 (95% CI: 1.0400–1.2013). Following the metric-conversion coefficient estimation, 10-ppbV by 6mMDA1 was identical to 8.31-ppbV by 6mDMA8 (**Figure 2**), and thus HR should be scaled to 1.1435 (95% CI: 1.0483–1.2470) for every 10-ppbV incremental exposure by 6mDMA8.

Supplementary Tables and Figures

Table S1 Searching strategy.

#21	#17 AND #18 AND #19 AND #20
#20	#14 OR #15 OR #16
#19	#11 OR #12 OR #13
#18	#3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10
#17	#1 OR #2
#16	(ozone OR O3 OR "O 3" OR "O(3)" OR "O-3" OR "air pollution").TW.
#15	(ozone OR O3 OR "O 3" OR "O(3)" OR "O-3" OR "air pollution").NM.
#14	(ozone OR "air pollution").SH.
#13	(long-term OR "long term").TW.
#12	(cohort OR Cox OR hazard OR HR OR risk OR prospective OR retrospective).TW.
#11	(cohort studies).SH.
#10	(lung cancer).TW.
#9	("cerebrovascular disease" OR stroke).TW.
#8	("congestive heart failure" OR CHF).TW.
#7	("ischemic heart disease" OR "ischaemic heart disease" OR IHD).TW.
#6	(cardiovascular OR CVD).TW.
#5	(respiratory OR "chronic obstructive pulmonary disease" OR COPD OR pneumonia).TW.
#4	(cardiopulmonary disease).TW.
#3	(all-cause OR "all cause").TW.
#2	(mortality OR death OR "premature death").TW.
#1	(mortality OR death OR "premature death").SH.

Search timeline: 1 September 2015 to 1 February 2022
SH: MeSH subject headings. NM: Name of substance word. TW: Text word.

Table S2 Quality assessment tool for observational cohort and cross-sectional studies.

A. Was the research question or objective in this paper clearly stated?
B. Was the study population clearly specified and defined?
C. Was the participation rate of eligible persons at least 50%?
D. Were all the subjects selected or recruited from the same or similar populations (including the same time period)? Were inclusion and exclusion criteria for being in the study prespecified and applied uniformly to all participants?
E. Was a sample size justification, power description, or variance and effect estimates provided?
F. For the analyses in this paper, were the exposure(s) of interest measured prior to the outcome(s) being measured?
G. Was the timeframe sufficient so that one could reasonably expect to see an association between exposure and outcome if it existed?
H. For exposures that can vary in amount or level, did the study examine different levels of the exposure as related to the outcome (e.g., categories of exposure, or exposure measured as continuous variable)?
I. Were the exposure measures (independent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?
J. Was the exposure(s) assessed more than once over time?
K. Were the outcome measures (dependent variables) clearly defined, valid, reliable, and implemented consistently across all study participants?
L. Were the outcome assessors blinded to the exposure status of participants?
M. Was loss to follow-up after baseline 20% or less?
N. Were key potential confounding variables measured and adjusted statistically for their impact on the relationship between exposure(s) and outcome(s)?

Source: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools.

Table S3 Adjusted key confounders of each included study.

Study	Key Confounding Adjustment
Abbey et al. 1999[2]	age, sex, BMI, smoking, individual demographic features‖, lifestyle features⊥, medical history
Lipfert et al. 2006[3]	age, ethnicity, BMI, smoking, traffic density, NO_2, CO
Jerrett et al. 2009[4]	
Krewski et al. 2009[5]	age, sex, ethnicity, BMI, smoking, individual demographic features, lifestyle features, $PM_{2.5}$
Smith et al. 2009[6]	
Lipsett et al. 2011[7]	age, ethnicity, BMI, smoking, lifestyle features, medical treatment
Zanobetti et al. 2011[8]	age, sex, ethnicity, medical history
Carey et al. 2013[9]	age, sex, BMI, smoking, individual demographic features
Jerrett et al. 2013[10]	age, sex, smoking, individual demographic features, lifestyle features
Bentayeb et al. 2015[11]	age, sex, BMI, smoking, individual demographic features, lifestyle features
Crouse et al. 2015[12]	age, sex, individual and area-level demographic features, $PM_{2.5}$, NO_2
Tonne et al. 2016[13]	age, sex, ethnicity, smoking, medical history, area-level demographic features
Turner et al. 2016[14]	age, sex, BMI, smoking, individual and area-level demographic features, $PM_{2.5}$, NO_2
Di et al. 2017[15]	age, sex, ethnicity, BMI, smoking, individual and area-level demographic features, meteorological features, $PM_{2.5}$
Weichenthal et al. 2017[16]	age, sex, ethnicity, individual and area-level demographic features
Cakmak et al. 2018[17]	age, sex, individual demographic features, $PM_{2.5}$
Hvidtfeldt et al. 2019[18]	age, sex, BMI, smoking, individual and area-level demographic features, noise
Kazemiparkouhi et al. 2019[20]	age, sex, ethnicity, area-level demographic features, $PM_{2.5}$
Lim et al. 2019[19]	age, sex, ethnicity, BMI, smoking, individual demographic features, $PM_{2.5}$, NO_2, daily maximum temperature
Paul et al. 2020[21]	age, sex, area-level demographic features
Shi et al. 2021[22]	age, sex, ethnicity, BMI, smoking, individual and area-level demographic features, lifestyle features, $PM_{2.5}$, NO_2, medical history
Strak et al. 2021[23]	age, sex, ethnicity, BMI, smoking, individual and area-level demographic features, $PM_{2.5}$, NO_2, BC
Yazdi et al. 2021[24]	age, sex, ethnicity, BMI, smoking, individual and area-level demographic features, lifestyle features, $PM_{2.5}$, NO_2, medical history
Bauwelinck et al. 2022[25]	age, sex, individual and area-level demographic features, $PM_{2.5}$, NO_2
Stafoggia et al. 2022[1]	age, sex, ethnicity, BMI, smoking, individual and area-level demographic features, $PM_{2.5}$, NO_2, BC

‖ Demographic features included marital status, education attainment, employment status and occupational class, aboriginal ancestry, visible minority ethnicities, immigrant status and residence location (urban or rural), income level and socioeconomic status (SES), and regional population density. Different studies adjusted various combinations of demographic features.

⊥ Lifestyle features included consumptions of alcohol, dietary fat, vegetables, fruits (dietary fibre), and vitamins, together with physical activity frequency. Different studies adjusted various combinations of lifestyle features.

Table S4 Subgroup meta-analyses of O_3 exposure-associated all-cause, respiratory and cardiovascular mortality risks. Meta-analysis for male and female populations were conducted based on the gender-specific risk association values directly reported in literature. Subgroups by study region, exposure metric, and methodology reliability rating of the individual exposure assignment (**Table 2**) were classified for all selected studies. The rest subgroups were defined binary based on whether the confounders were adjusted in risk estimation instead of using the reported grouped risk estimations, due to inconsistent categorisation criteria across studies and information availability. The significances of inter-group differences were indicated by *p-values*.

		All-cause Mortality		**Respiratory Mortality**		**Cardiovascular Mortality**	
		RR	*p-value*	RR	*p-value*	RR	*p-value*
Subgroup Categorisation							
Sex	*Male*	1.036 (1.027, 1.044)	0.19	1.036 (1.018, 1.054)	0.16	-	-
	Female	1.025 (1.011, 1.039)		1.017 (0.997, 1.037)		-	
Region	*North America*	1.019 (1.014, 1.024)	0.02	1.029 (1.011, 1.047)	0.07	1.036 (1.017, 1.056)	0.01
	Europe	0.910 (0.827, 1.001)		0.941 (0.856, 1.036)		0.934 (0.866, 1.008)	
Metric	*6mDMA1*	1.004 (0.991, 1.017)	0.001	1.041 (1.033, 1.049)	<0.0001	0.993 (0.986, 1.000)	0.01
	6mDMA8	1.022 (1.014, 1.030)		1.000 (0.961, 1.040)		1.050 (1.012, 1.090)	
	ADA24	0.980 (0.960, 1.001)		0.972 (0.929, 1.018)		0.999 (0.983, 1.015)	
Methodology Reliability	*Low*	1.002 (0.993, 1.011)	0.002	1.038 (1.033, 1.043)	0.003	0.998 (0.992, 1.004)	<0.0001
	Good	1.042 (1.007, 1.077)		1.010 (0.990, 1.030)		1.099 (1.037, 1.164)	
	High	1.017 (1.008, 1.025)		1.058 (1.037, 1.079)		1.021 (1.004, 1.038)	
Confounder Adjustment							
Ethnicity	*Yes*	1.015 (1.009, 1.020)	0.25	1.025 (1.011, 1.040)	0.49	1.026 (1.007, 1.045)	0.93
	No	0.999 (0.972, 1.026)		1.005 (0.953, 1.061)		1.027 (0.997, 1.059)	
BMI	*Yes*	1.008 (1.001, 1.015)	0.11	1.002 (0.961, 1.044)	0.29	0.998 (0.979, 1.017)	0.09
	No	1.025 (1.005, 1.046)		1.026 (1.012, 1.040)		1.059 (1.008, 1.112)	
Smoking History	*Yes*	1.007 (1.001, 1.014)	0.06	1.002 (0.963, 1.042)	0.29	1.001 (0.985, 1.016)	0.07
	No	1.034 (1.007, 1.062)		1.029 (0.999, 1.059)		1.069 (1.002, 1.140)	
Lifestyle Features	*Yes*	1.014 (0.990, 1.039)	0.60	1.033 (1.006, 1.061)	0.14	1.002 (0.985, 1.020)	0.14
	No	1.007 (0.999, 1.016)		1.000 (0.968, 1.034)		1.032 (0.997, 1.069)	
$PM_{2.5}$	*Yes*	1.014 (1.009, 1.020)	0.15	1.031 (1.000, 1.063)	0.14	1.014 (0.997, 1.031)	0.46
	No	0.993 (0.965, 1.021)		0.993 (0.954, 1.033)		1.040 (0.974, 1.110)	
NO_2	*Yes*	0.996 (0.968, 1.025)	0.43	0.999 (0.941, 1.061)	0.49	1.024 (1.007, 1.042)	0.84
	No	1.008 (1.001, 1.015)		1.022 (0.997, 1.047)		1.029 (0.992, 1.067)	

Table S5 Quality assessment of 25 included cohort studies for meta-analysis.

Study	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Score
Abbey et al. 1999[2]	√	√	√	√	√	√	√	√		√	√	√	√	√	Fair
Lipfert et al. 2006[3]	√	√	√		√	√	√	√		√	√	√	√	√	Fair
Jerrett et al. 2009[4]	√	√	√	√	√	√	√	√		√	√	√	√	√	Fair
Krewski et al. 2009[5]	√	√	√	√	√		√	√		√	√	√	√	√	Fair
Smith et al. 2009[6]	√	√	√		√		√	√		√	√	√	√	√	Fair
Lipsett et al. 2011[7]	√	√	√		√	√	√	√		√	√	√	√	√	Fair
Zanobetti et al. 2011[8]	√	√	√	√	√	√	√	√		√	√	√	√	√	Fair
Carey et al. 2013[9]	√	√	√		√	√		√		√	√	√	√	√	Fair
Jerrett et al. 2013[10]	√	√	√	√	√	√	√	√		√	√	√	√	√	Fair
Bentayeb et al. 2015[11]	√	√	√		√	√	√	√	√	√	√	√	√	√	Fair
Crouse et al. 2015[12]	√	√	√		√	√	√	√	√	√	√	√	√	√	Fair
Tonne et al. 2016[13]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Turner et al. 2016[14]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Di et al. 2017[15]	√	√	√		√	√	√	√	√	√	√	√	√	√	Fair
Weichenthal et al. 2017[16]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Cakmak et al. 2018[17]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Hvidtfeldt et al. 2019[18]	√	√	√		√	√		√	√	√	√	√	√	√	Fair
Kazemiparkouhi et al. 2019[20]	√	√	√		√	√	√	√		√	√	√	√	√	Fair
Lim et al. 2019[19]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Paul et al. 2020[21]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Shi et al. 2021[22]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Strak et al. 2021[23]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Yazdi et al. 2021[24]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Bauwelinck et al. 2022[25]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good
Stafoggia et al. 2022[1]	√	√	√	√	√	√	√	√	√	√	√	√	√	√	Good

Table S6 **GRADE assessment for epidemiological evidences on the association between long-term O_3 exposure and cause-specific mortality risks.**

Mortality Cause: All Causes		
Domains	**Assessment**	**Downgrade/Upgrade**
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.81 to 1.11.	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, and 3 out of 23 studies (Di et al. 2017; Shi et al. 2021; Stafoggia et al. 2022) checked the dose-response trends.	Upgrade
Plausible Confounding towards Null	Cakmak et al. 2018 reported higher RR after adjusting confounders.	Upgrade
Overall Judgment	**High**	

Mortality Cause: All Respiratory Diseases		
Domains	**Assessment**	**Downgrade/Upgrade**
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.75 to 1.17	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, and 2 out of 16 studies (Jerrett et al. 2009; Lim et al. 2019) tested the dose-response trends.	Upgrade
Plausible Confounding towards Null	No crude and adjusted risks were provided for each study.	No change
Overall Judgment	**High**	

Mortality Cause: Chronic Obstructive Pulmonary Disease (COPD)		
Domains	**Assessment**	**Downgrade/Upgrade**
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.74 to 1.15.	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, but no studies considered dose-response trends.	No change
Plausible Confounding towards Null	No crude and adjusted risks were provided for each study.	No change
Overall Judgment	**Moderate**	

Table S6 (cont'd) **GRADE assessment for epidemiological evidences on the association between long-term O_3 exposure and cause-specific mortality risks.**

Mortality Cause: Cardiovascular Diseases (CVD)

Domains	Assessment	Downgrade/Upgrade
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.83 to 1.16.	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, and 2 studies (Lim et al. 2019; Paul et al. 2020) checked the dose-response trends.	Upgrade
Plausible Confounding towards Null	No crude and adjusted risks were provided for each study.	No change
Overall Judgment	**High**	

Mortality Cause: Ischaemic Heart Disease (IHD)

Domains	Assessment	Downgrade/Upgrade
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.76 to 1.12.	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, but no studies considered dose-response trends.	No change
Plausible Confounding towards Null	Cakmak et al. 2018 reported higher RR after adjusting confounders.	Upgrade
Overall Judgment	**High**	

Mortality Cause: Lung Cancer (LC)

Domains	Assessment	Downgrade/Upgrade
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.81 to 1.71.	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, but no studies considered dose-response trends.	No change
Plausible Confounding towards Null	Cakmak et al. 2018 reported higher RR after adjusting confounders.	Upgrade
Overall Judgment	**High**	

Table S6 (cont'd) **GRADE assessment for epidemiological evidences on the association between long-term O_3 exposure and cause-specific mortality risks.**

Mortality Cause: Cerebrovascular Disease (CEVD)		
Domains	**Assessment**	**Downgrade/Upgrade**
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were inconsistent, as the point estimates were in the range of 0.78 to 1.03.	Downgrade
Indirectness	All studies included the desired population, exposures and outcomes.	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, but no studies considered dose-response trends.	No change
Plausible Confounding towards Null	No crude and adjusted risks were provided for each study.	No change
Overall Judgment	**Moderate**	
Mortality Cause: Congestive Heart Failure (CHF)		
Domains	**Assessment**	**Downgrade/Upgrade**
Start Level	All cohort studies.	High
Risk of Bias	The overall risk of bias in all cohorts was low.	No change
Imprecision	All studies included reported the 95% confidence interval around the best estimate of the absolute effect.	No change
Inconsistency	The values of effect sizes across the studies were consistent, as the point estimates were in the range of 1.01 to 1.26.	No change
Indirectness	All studies included the desired population, exposures and outcomes	No change
Publication Bias	Given the comprehensive search, there is no need to consider publication bias.	No change
Magnitude of Associations	The magnitude of effect sizes was not large enough to upgrade the level of evidence.	No change
Dose-Response Trend	Linear dose-response relationship was assumed in all studies, but no studies considered dose-response trends.	No change
Plausible Confounding towards Null	No crude and adjusted risks were provided for each study.	No change
Overall Judgment	**High**	

Table S7 Leave-one-out sensitivity analysis of long-term O_3 exposure-associated all-cause, respiratory diseases, COPD, cardiovascular diseases, congestive heart failure, ischaemic heart disease, cerebrovascular diseases, and lung cancer mortality risks by 6mDMA8 exposure metric. The pooled relative risks (RR), 95% confidence intervals (95% CI) presented in lower bounds (LB) and upper bounds (UB), and heterogeneity I^2 were estimated after excluding each study.

Study	RR	LB	UB	I^2
All-cause Mortality				
Abbey et al. 1999 (male cohort)	1.014	1.009	1.019	97.7%
Abbey et al. 1999 (female cohort)	1.014	1.009	1.019	97.7%
Lipfert et al. 2006	1.013	1.008	1.018	97.7%
Jerrett et al. 2009	1.016	1.011	1.021	97.7%
Krewski et al. 2009	1.013	1.008	1.018	97.7%
Smith et al. 2009	1.014	1.009	1.019	97.8%
Lipsett et al. 2011	1.016	1.011	1.021	97.7%
Carey et al. 2013	1.014	1.009	1.019	97.7%
Jerrett et al. 2013	1.015	1.010	1.020	97.7%
Bentayeb et al. 2015	1.014	1.009	1.019	97.7%
Crouse et al. 2015	1.014	1.009	1.019	97.7%
Tonne et al. 2016	1.014	1.009	1.019	97.8%
Turner et al. 2016	1.014	1.008	1.019	97.7%
Di et al. 2017	1.011	1.005	1.018	97.7%
Weichenthal et al. 2017	1.011	1.006	1.016	97.5%
Cakmak et al. 2018	1.014	1.009	1.018	97.7%
Hvidtfeldt et al. 2019	1.015	1.010	1.020	97.7%
Kazemiparkouhi et al. 2019	1.014	1.008	1.020	97.5%
Lim et al. 2019	1.015	1.010	1.020	97.7%
Shi et al. 2021	1.008	1.004	1.012	95.3%
Strak et al. 2021	1.017	1.012	1.022	97.4%
Yazdi et al. 2021	1.009	1.001	1.018	97.8%
Bauwelinck et al. 2022	1.013	1.008	1.018	97.7%
Stafoggia et al. 2022	1.015	1.010	1.020	97.7%
Respiratory Mortality				
Abbey et al. 1999 (male cohort)	1.027	1.013	1.040	84.3%
Abbey et al. 1999 (female cohort)	1.027	1.014	1.040	84.4%
Jerrett et al. 2009	1.025	1.011	1.039	84.3%
Smith et al. 2009	1.025	1.012	1.038	83.6%
Lipsett et al. 2011	1.028	1.014	1.042	84.1%
Carey et al. 2013	1.031	1.019	1.043	78.9%
Jerrett et al. 2013	1.030	1.016	1.044	83.2%
Bentayeb et al. 2015	1.027	1.014	1.040	84.4%
Crouse et al. 2015	1.033	1.020	1.046	81.2%
Turner et al. 2016	1.020	1.007	1.034	81.6%
Weichenthal et al. 2017	1.025	1.011	1.039	84.4%
Hvidtfeldt et al. 2019	1.028	1.015	1.042	84.0%
Kazemiparkouhi et al. 2019	1.012	0.983	1.041	83.3%
Lim et al. 2019	1.026	1.012	1.040	84.4%
Strak et al. 2021	1.029	1.016	1.042	82.4%
Bauwelinck et al. 2022	1.025	1.011	1.039	84.2%
Stafoggia et al. 2022	1.030	1.017	1.043	82.1%
COPD Mortality				
Zanobetti et al. 2011	1.041	1.013	1.071	95.2%
Crouse et al. 2015	1.075	1.050	1.100	91.0%
Turner et al. 2016	1.042	1.013	1.073	95.3%
Cakmak et al. 2018	1.067	1.040	1.095	93.0%
Kazemiparkouhi et al. 2019	1.031	0.973	1.093	90.8%
Lim et al. 2019	1.052	1.022	1.082	95.2%
Strak et al. 2021	1.058	1.032	1.086	94.7%
Cardiovascular Mortality				
Jerrett et al. 2009	1.031	1.015	1.048	97.8%
Smith et al. 2009	1.025	1.009	1.041	97.8%
Lipsett et al. 2011	1.029	1.012	1.046	97.8%
Jerrett et al. 2013	1.028	1.012	1.045	97.8%
Bentayeb et al. 2015	1.027	1.011	1.043	97.8%
Crouse et al. 2015	1.025	1.009	1.042	97.7%
Turner et al. 2016	1.027	1.010	1.044	97.8%
Weichenthal et al. 2017	1.015	1.005	1.025	92.3%
Hvidtfeldt et al. 2019	1.033	1.017	1.049	97.7%
Kazemiparkouhi et al. 2019	1.027	0.996	1.059	96.9%
Lim et al. 2019	1.027	1.011	1.044	97.8%
Paul et al. 2020	1.021	1.005	1.037	97.5%
Bauwelinck et al. 2022	1.025	1.009	1.041	97.7%
Stafoggia et al. 2022	1.031	1.015	1.047	97.8%
Congestive Heart Failure Mortality				
Zanobetti et al. 2011	1.079	1.051	1.107	87.7%
Turner et al. 2016	1.073	1.040	1.107	89.3%
Kazemiparkouhi et al. 2019	1.108	1.025	1.198	91.1%
Lim et al. 2019	1.103	1.079	1.129	78.9%
Ischaemic Heart Disease Mortality				
Jerrett et al. 2009	1.023	1.004	1.043	96.8%
Krewski et al. 2009	1.017	0.998	1.037	96.9%
Lipsett et al. 2011	1.016	0.997	1.036	96.9%
Jerrett et al. 2013	1.016	0.997	1.036	96.9%
Crouse et al. 2015	1.012	0.994	1.030	96.2%
Turner et al. 2016	1.021	1.001	1.040	96.9%
Cakmak et al. 2018	1.008	0.994	1.021	92.8%
Kazemiparkouhi et al. 2019	1.012	0.979	1.045	95.7%
Lim et al. 2019	1.015	0.996	1.035	96.9%
Strak et al. 2021	1.023	1.005	1.041	96.6%
Cerebrovascular Mortality				
Lipsett et al. 2011	0.991	0.972	1.010	82.2%
Crouse et al. 2015	0.986	0.970	1.002	77.0%
Turner et al. 2016	0.988	0.972	1.005	79.9%
Kazemiparkouhi et al. 2019	0.985	0.947	1.023	79.8%
Lim et al. 2019	0.996	0.980	1.012	79.0%
Strak et al. 2021	0.993	0.980	1.007	71.4%
Lung Cancer Mortality				
Abbey et al. 1999 (male cohort)	0.963	0.927	1.001	86.4%
Abbey et al. 1999 (female cohort)	0.966	0.901	1.035	86.9%
Krewski et al. 2009	0.962	0.892	1.036	86.8%
Lipsett et al. 2011	0.962	0.892	1.036	86.7%
Carey et al. 2013	0.975	0.914	1.041	85.4%
Jerrett et al. 2013	0.964	0.894	1.039	85.8%
Crouse et al. 2015	0.963	0.894	1.038	85.5%
Turner et al. 2016	0.964	0.894	1.039	85.9%
Cakmak et al. 2018	0.957	0.889	1.030	86.9%
Kazemiparkouhi et al. 2019*	0.958	0.890	1.032	62.7%
Lim et al. 2019	0.962	0.893	1.037	85.5%
Bauwelinck et al. 2022	0.968	0.898	1.042	85.0%
Stafoggia et al. 2022	0.974	0.908	1.045	84.0%

* Kazemiparkouhi et al. 2019 was highlighted due to the unneglectable change in the heterogeneity. The inter-study heterogeneities ranged 84.0-86.9% when eliminating any of the rest studies, while excluding Kazemiparkouhi et al. 2019 reduced the heterogeneity significantly to 62.7%.

Table S8 Pooled RRs for all-cause, respiratory, COPD, cardiovascular, ischaemic heart disease, and lung cancer mortality after excluding studies from duplicated cohorts. Only the latest studies with the largest population coverage from the same cohort for each mortality cause were included for more rigorous meta-analysis. The pooled RRs stated in the Result section considered all selected studies, following the 2 previous systematic reviews (Atkinson et al. 2016 and Huangfu et al. 2019). Hunter-Schmidt random-effects meta-regression estimator was applied for RR pooling. Mortality risks of congestive heart failure and all cerebrovascular diseases were no longer analysed as no cohorts were used more than once.

Study	Cohort	RR (95% CI)
All-cause Mortality		
Abbey et al. 1999	AHS-M	1.064 (0.964, 1.174)
Abbey et al. 1999	AHS-F	0.964 (0.890, 1.043)
Lipfert et al. 2006	WU-EPRI	1.033 (1.012, 1.053)
Lipsett et al. 2011	CTS	0.993 (0.986, 1.002)
Carey et al. 2013	CPRD	0.871 (0.782, 0.934)
Bentayeb et al. 2015	GAZEL	0.816 (0.646, 1.032)
Tonne et al. 2016	MINAP	0.962 (0.834, 1.098)
Turner et al. 2016	ACS CPS II	1.020 (1.010, 1.040)
Cakmak et al. 2018	CANCHEC	1.080 (1.020, 1.140)
Hvidtfeldt et al. 2019	DDCH	0.949 (0.908, 1.000)
Lim et al. 2019	NIH-AARP	1.000 (0.990, 1.010)
Yazdi et al. 2021	Medicare	1.008 (1.008, 1.008)
Bauwelinck et al. 2022	BC2001	1.036 (1.014, 1.058)
Stafoggia et al. 2022	ELAPSE	0.910 (0.866, 0.959)
Random-effects model		**1.008 (1.006, 1.009)**
Heterogeneity: $I^2 = 82.6\%$, $\tau^2 < 0.0001$, $p < 0.01$		
Respiratory Diseases		
Abbey et al. 1999	AHS-M	1.085 (0.890, 1.319)
Abbey et al. 1999	AHS-F	1.036 (0.867, 1.241)
Lipsett et al. 2011	CTS	1.020 (0.993, 1.044)
Carey et al. 2013	CPRD	0.782 (0.699, 0.871)
Bentayeb et al. 2015	GAZEL	0.953 (0.554, 1.671)
Turner et al. 2016	ACS CPS II	1.103 (1.069, 1.137)
Weichenthal et al. 2017	CANCHEC	1.041 (1.011, 1.070)
Hvidtfeldt et al. 2019	DDCH	0.970 (0.888, 1.051)
Kazemiparkouhi et al. 2019	Medicare	1.040 (1.036, 1.045)
Lim et al. 2019	NIH-AARP	1.040 (1.000, 1.080)
Bauwelinck et al. 2022	BC2001	1.062 (1.014, 1.111)
Stafoggia et al. 2022	ELAPSE	0.901 (0.831, 0.977)
Random-effects model		**1.034 (1.017, 1.050)**
Heterogeneity: $I^2 = 81.7\%$, $\tau^2 = 0.0002$, $p < 0.01$		
Chronic Pulmonary Obstructive Disease		
Turner et al. 2016	ACS CPS II	1.120 (1.069, 1.179)
Cakmak et al. 2018	CANCHEC	1.000 (0.970, 1.030)
Kazemiparkouhi et al. 2019	Medicare	1.102 (1.096, 1.108)
Lim et al. 2019	NIH-AARP	1.060 (1.010, 1.120)
Strak et al. 2021	ELAPSE	0.746 (0.605, 0.917)
Random-effects model		**1.060 (1.040, 1.080)**
Heterogeneity: $I^2 = 90.2\%$, $\tau^2 = 0.0002$, $p < 0.01$		
Cardiovascular Diseases		
Lipsett et al. 2011	CTS	1.004 (0.991, 1.015)
Bentayeb et al. 2015	GAZEL	0.831 (0.397, 1.729)
Turner et al. 2016	ACS CPS II	1.026 (1.009, 1.043)
Weichenthal et al. 2017	CANCHEC	1.161 (1.144, 1.178)
Hvidtfeldt et al. 2019	DDCH	0.936 (0.902, 0.979)
Kazemiparkouhi et al. 2019	Medicare	0.996 (0.994, 0.999)
Lim et al. 2019	NIH-AARP	1.020 (0.990, 1.030)
Paul et al. 2020	ONPHEC	1.105 (1.078, 1.133)
Bauwelinck et al. 2022	BC2001	1.050 (1.022, 1.076)
Stafoggia et al. 2022	ELAPSE	0.954 (0.912, 0.996)
Random-effects model		**1.032 (1.010, 1.055)**
Heterogeneity: $I^2 = 98.2\%$, $\tau^2 = 0.0009$, $p < 0.01$		
Ischaemic Heart Disease		
Lipsett et al. 2011	CTS	1.020 (1.002, 1.040)
Turner et al. 2016	ACS CPS II	0.983 (0.957, 1.000)
Cakmak et al. 2018	CANCHEC	1.120 (1.100, 1.130)
Kazemiparkouhi et al. 2019	Medicare	0.995 (0.992, 0.999)
Lim et al. 2019	NIH-AARP	1.030 (1.000, 1.060)
Strak et al. 2021	ELAPSE	0.761 (0.679, 0.851)
Random-effects model		**1.008 (0.973, 1.045)**
Heterogeneity: $I^2 = 99.2\%$, $\tau^2 = 0.0017$, $p < 0.01$		
Lung Cancer		
Abbey et al. 1999	AHS-M	1.705 (0.993, 2.921)
Abbey et al. 1999	AHS-F	0.829 (0.489, 1.408)
Lipsett et al. 2011	CTS	0.989 (0.956, 1.022)
Carey et al. 2013	CPRD	0.811 (0.699, 0.934)
Turner et al. 2016	ACS CPS II	0.965 (0.922, 1.000)
Cakmak et al. 2018	CANCHEC	1.040 (0.970, 1.120)
Kazemiparkouhi et al. 2019	Medicare	1.019 (1.013, 1.025)
Lim et al. 2019	NIH-AARP	0.980 (0.950, 1.000)
Bauwelinck et al. 2022	BC2001	0.927 (0.880, 0.975)
Stafoggia et al. 2022	ELAPSE	0.857 (0.791, 0.929)
Random-effects model		**0.966 (0.931, 1.002)**
Heterogeneity: $I^2 = 83.8\%$, $\tau^2 = 0.0004$, $p < 0.01$		

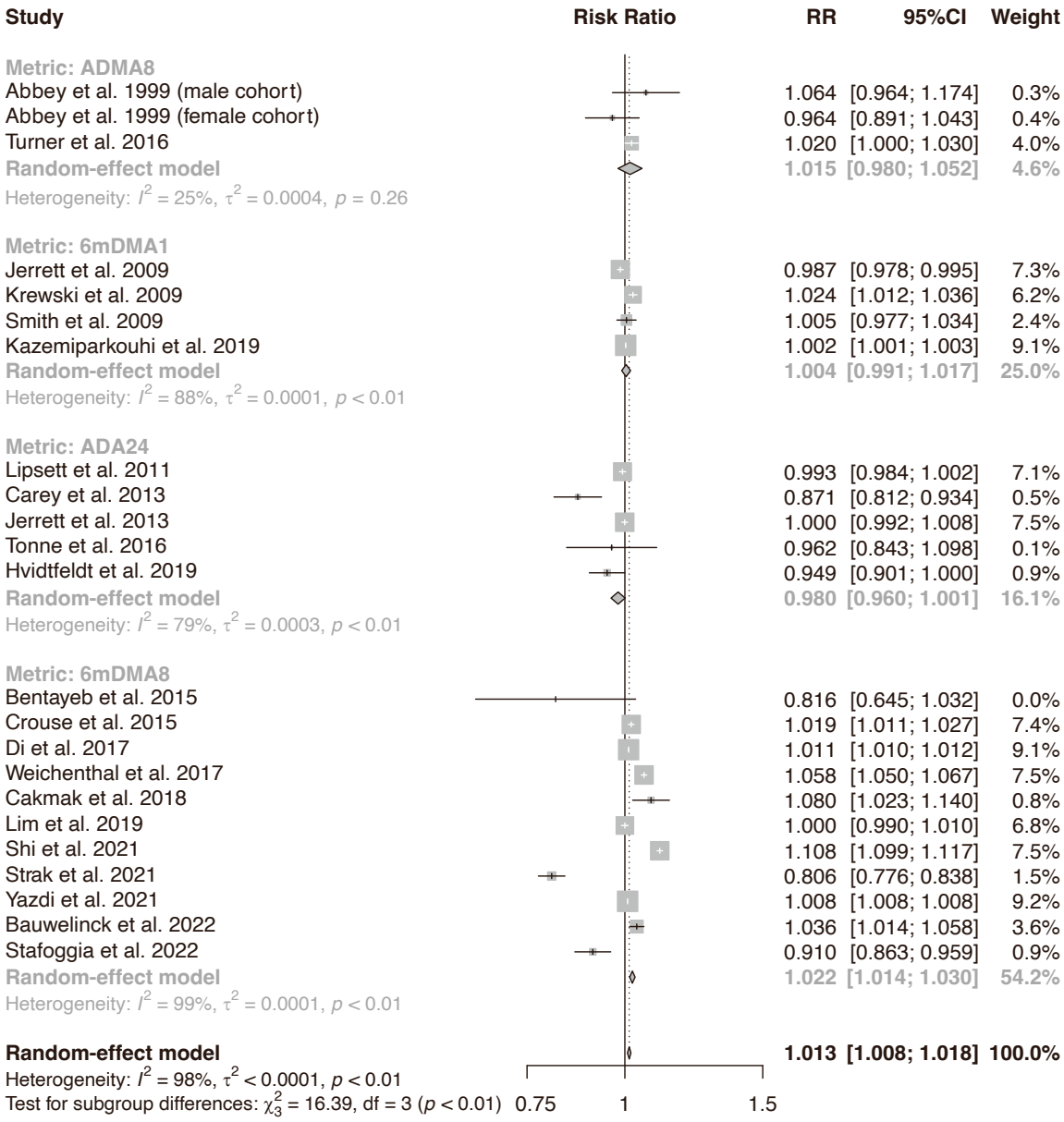


Figure S1 Sub-group meta-analyses by metrics of non-accidental all-cause mortality risks attributable to every 10-ppbV incremental long-term O_3 exposure. The overall pooled RR was slightly different from the value given in **Table 3**, as the only study, Lipfert et al. 2006, using exposure metric as ADMA1, was excluded from meta-analysis.

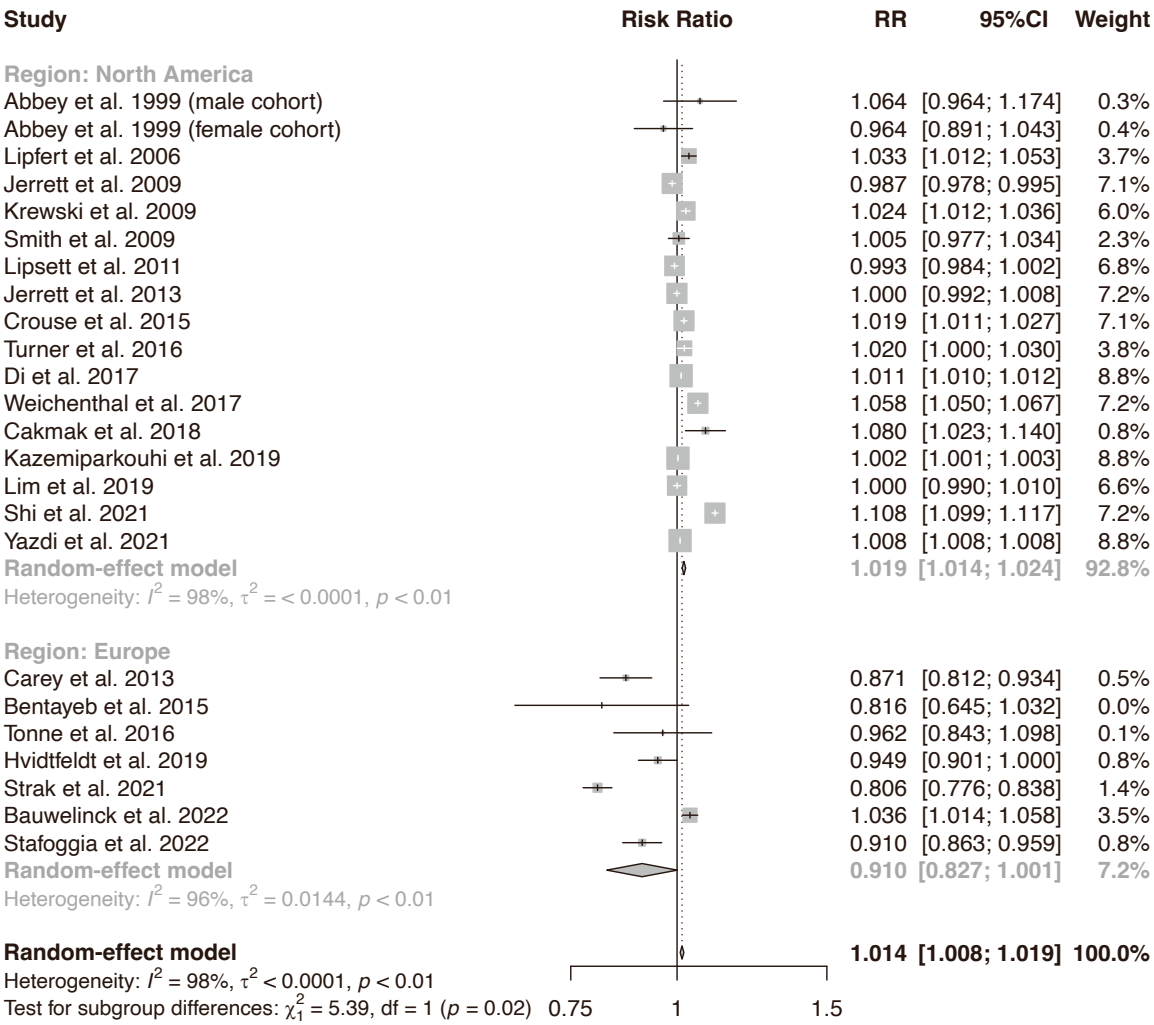


Figure S2 Sub-group meta-analyses by cohort population ethnicity of non-accidental all-cause mortality risks attributable to every 10-ppbV incremental long-term O_3 exposure.

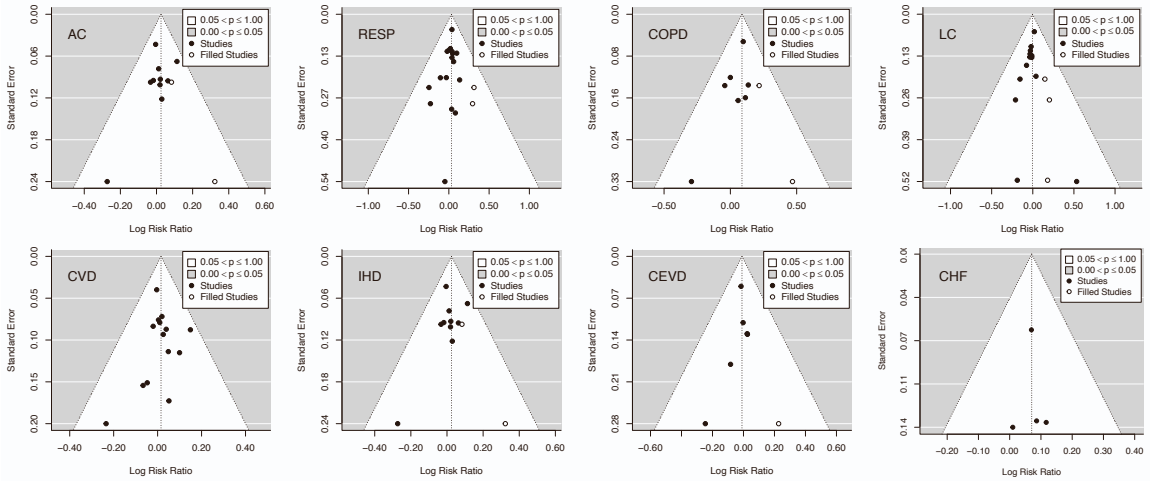


Figure S3 Publication bias test by trim-and-fill method for 8 causes of O_3-exposure attributable mortality risks. Biases were highlighted by statistically filled values to reach symmetry between risks higher and lower than pooled values.

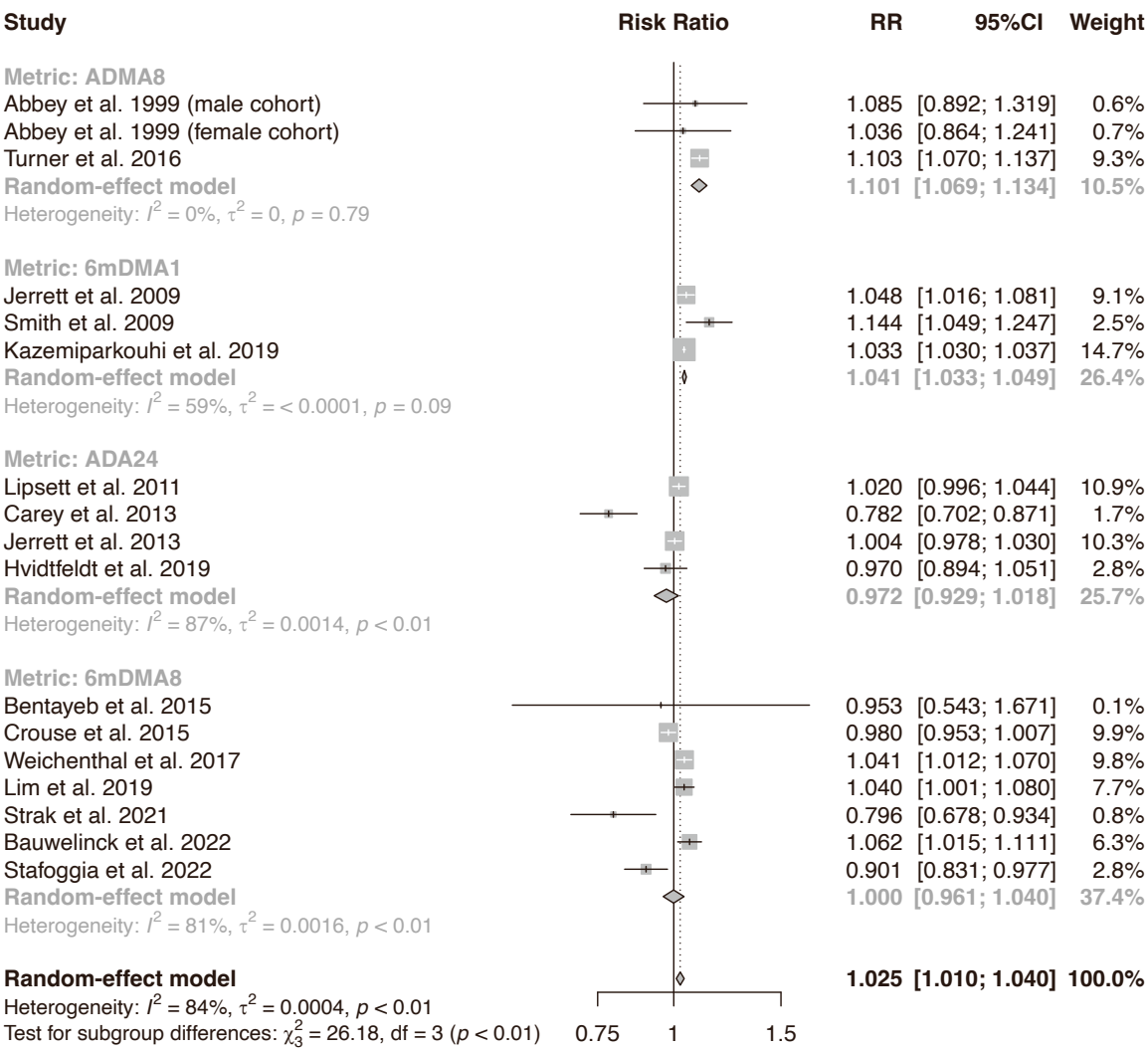


Figure S4 Sub-group meta-analyses by metrics of all respiratory mortality risks attributable to every 10-ppbV incremental long-term O_3 exposure.

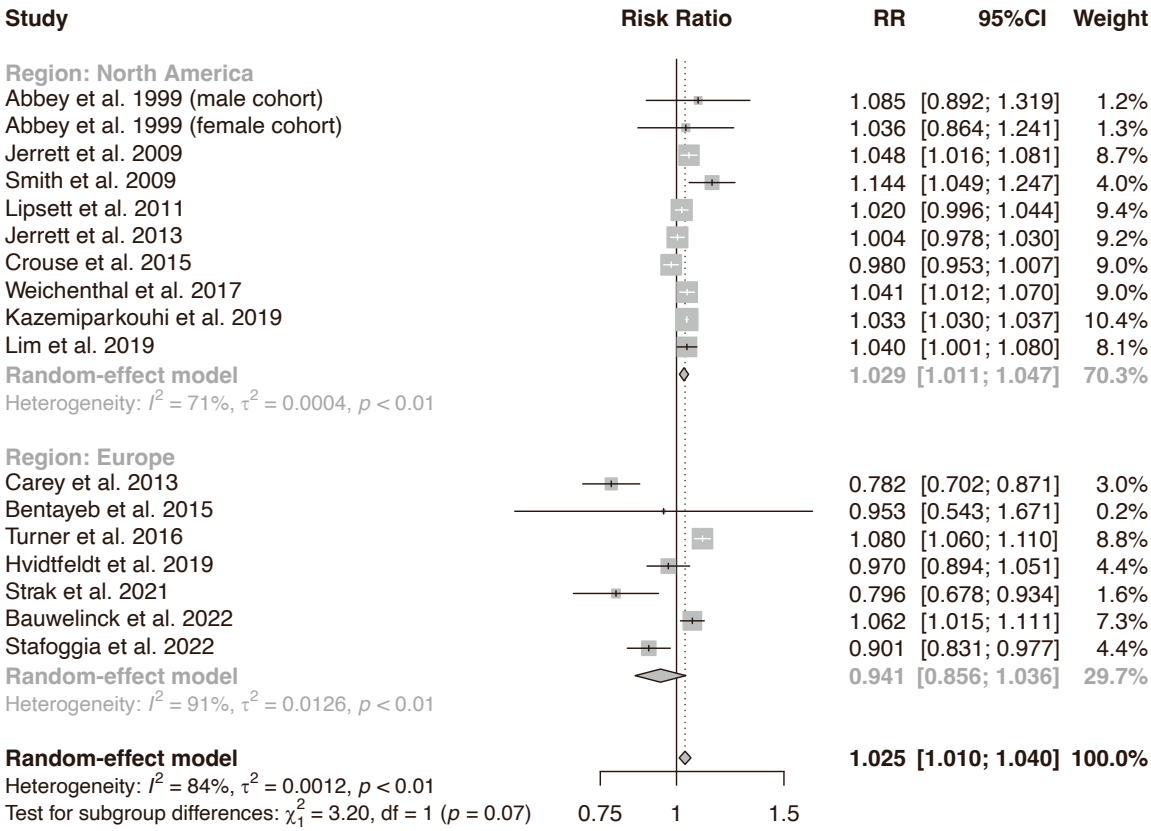


Figure S5 Sub-group meta-analyses by cohort population ethnicity of all respiratory mortality risks attributable to every 10-ppbV incremental long-term O_3 exposure.

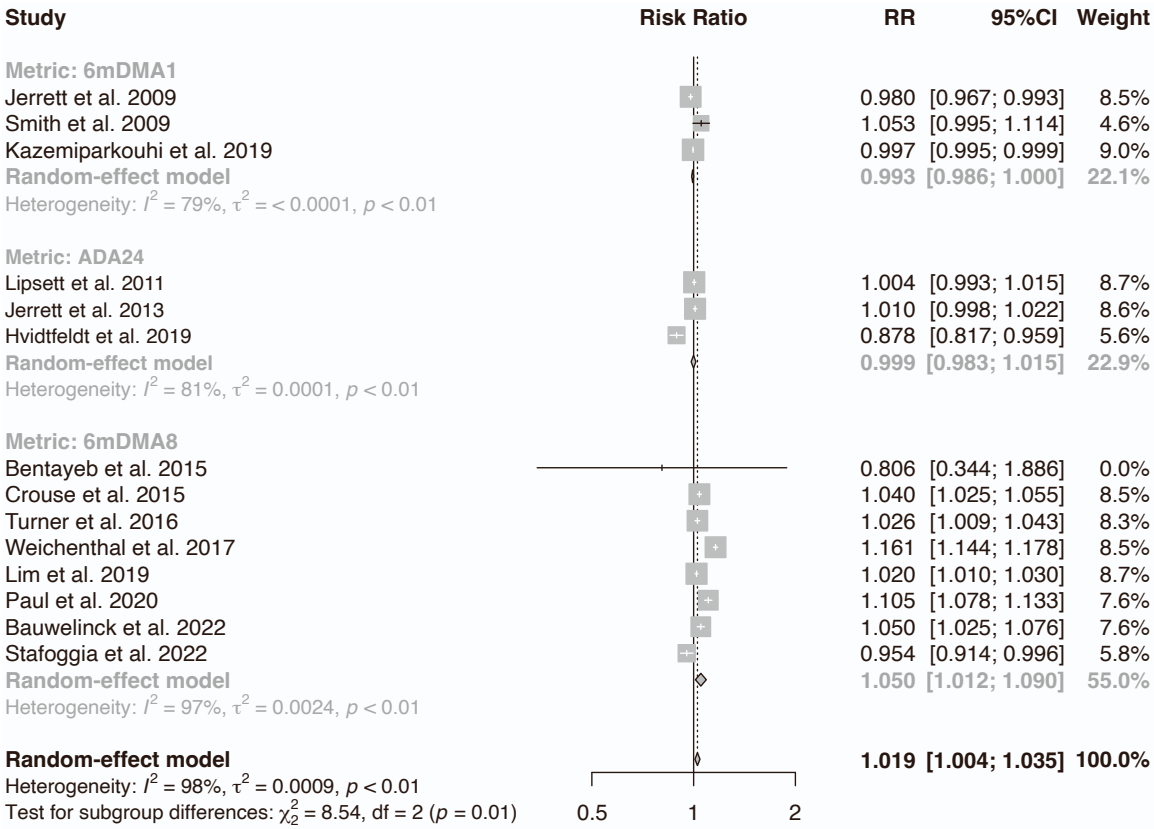


Figure S6 Sub-group meta-analyses by metrics of cardiovascular mortality risks attributable to every 10-ppbV incremental long-term O_3 exposure.

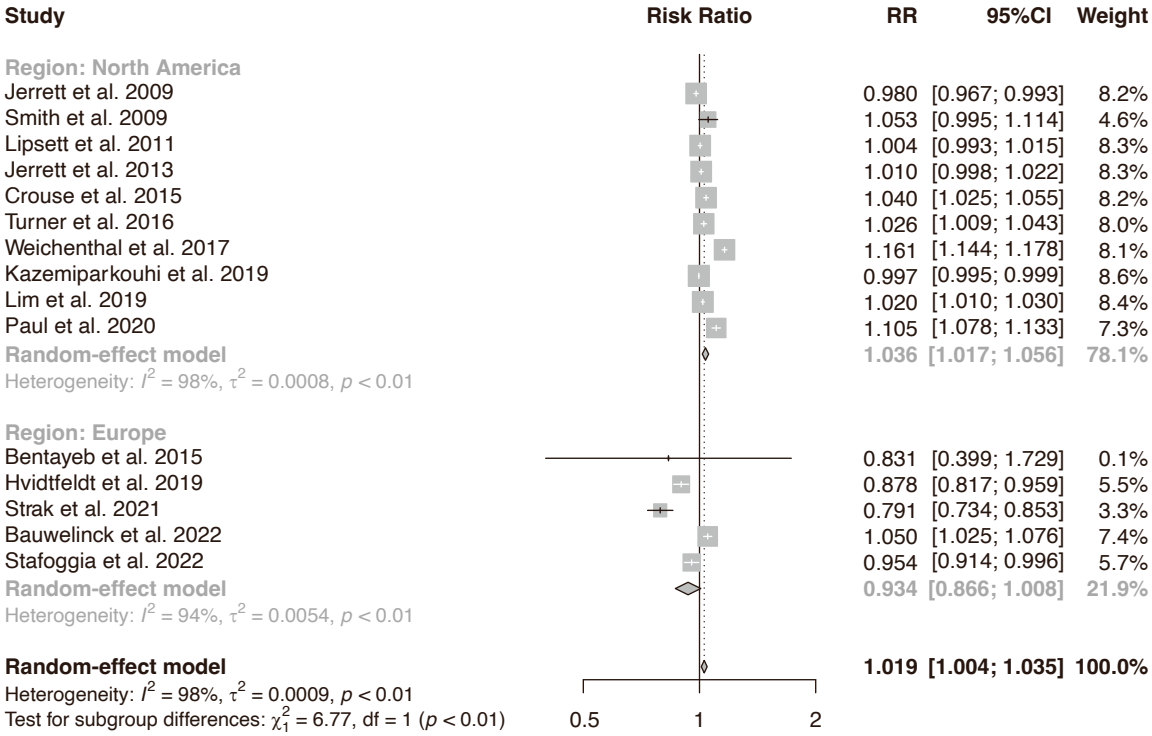


Figure S7 Sub-group meta-analyses by cohort population ethnicity of cardiovascular mortality risks attributable to every 10-ppbV incremental long-term O_3 exposure.

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